

Latent Outlet Potential & Potential-Based Trade Spend Allocation

Data Storm v7.0 – Final Round

Sri Lanka · January 2026 horizon

Submitted by Smile Labs

Powered by OCTAVE – John Keells Group | Rotaract Club of University of Moratuwa

One-line method

We estimate $\mathbb{E}[\text{true_demand}_i \mid X_i]$ where observed sales $y_i = \min(\text{true_demand}_i, \text{constraint}_i)$ are *right-censored*. The estimand is *not point-identified* from observational data alone, so we report a [lower, point, upper] band per outlet – the point combining Stochastic Frontier Analysis, multi-quantile gradient-boosted regression, Chernozhukov–Hong censoring correction, Conformalised QR, and bootstrap peer-bucket caps. On top of these predictions we add a **decision layer**: a Lagrangian water-filling optimizer for the LKR 5M Western budget, a functional LLM-based Explainable-AI module, and an interactive Outlet Intelligence web app.

Headline	Value
Outlets predicted (full population in scope)	20,000
External POIs scraped (Geofabrik PBF, 9 categories)	42,386
Outlets with ≥ 1 POI within 2 km	80.9 %
Rejected check records (coords + txn + holidays)	10,179 / 3 datasets
Median uplift vs. historical max	1.250×
Submission preflight (6 automated checks)	6 / 6 PASS
LKR 5M Western budget – outlets funded	3,729 / 9,000
Expected incremental volume (Western)	+109,021 L
Revenue-to-spend ROI	5.53×

Table 1: Headline value across prediction and decision layers.

Build trail

The modeling stack was built and audited across **seven rounds of AI council reviews** (four parallel premium-model critics per round). Every fix in `src/` cites the council finding it addresses (**FIX R7**, **FIX R4**, **FIX M1**, ...). Provenance: `Reviews/` and `Docs/ai_transparency_log.md`.

1 Data Engineering and Scraping Pipeline

1.1 Bronze → Silver → Gold lakehouse

Raw CSVs ingest *as-is* into `data/bronze/` with a SHA-256 audit log (idempotent: re-runs overwrite deterministically and re-hash). The **Silver** layer applies six reusable, parameterizable data-quality functions (`src/quality/checks.py`) consistently across all five raw datasets; failed rows are quarantined to `data/silver_rejected/` carrying `dataset_name`, `failed_check`, and `failure_reason`. **Gold** is the model-ready outlet × feature parquet.

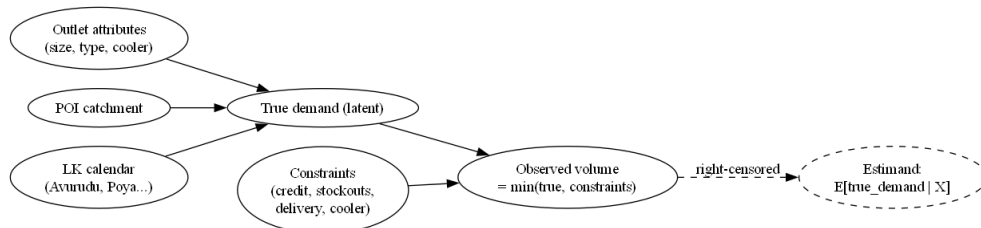


Figure 1: Bronze → Silver → Gold pipeline and the right-censoring DAG: observed sales are the min of latent demand and constraint, so the estimand is only partially identified.

1.2 POI acquisition: Geofabrik PBF » naïve Overpass

Naïvely querying Overpass for $20,000 \times 9 \text{ cats} \times 4 \text{ radii} = 720,000$ requests is infeasible (timeouts + ToS). We instead download the **Geofabrik Sri Lanka extract** (`sri-lanka-latest.osm.pbf`, ~136 MB) once, parse locally with `pyosm`, and run nearest-neighbour search via `BallTree(metric="haversine")`. End-to-end 25–45 min, offline after one fetch. Lives in `poi_pipeline/` as an idempotent sub-project.

Category	POIs	OSM tags
Schools / universities	5,694	<code>amenity</code> ∈ {school, university, college}
Transport hubs	5,799	<code>highway=bus_stop</code> , <code>railway=station</code>
Hospitals + healthcare	2,007	<code>amenity</code> ∈ {hospital, clinic, pharmacy}
Restaurants / cafes	4,667	<code>amenity</code> ∈ {restaurant, cafe, fast_food}
Supermarkets / groceries	2,543	<code>shop</code> ∈ {supermarket, convenience}
Religious places	9,255	<code>amenity=place_of_worship</code>
Hotels / attractions	5,513	<code>tourism</code> ∈ {hotel, guest_house}
Offices	4,169	<code>office=*</code> , <code>amenity=townhall</code>
Banks / ATMs	2,739	<code>amenity</code> ∈ {bank, atm}
Total	42,386	80.9% outlets with ≥ 1 POI within 2 km

Table 2: POI extraction summary (`poi_pipeline/data/pois/`).

1.3 Features for footfall and market potential

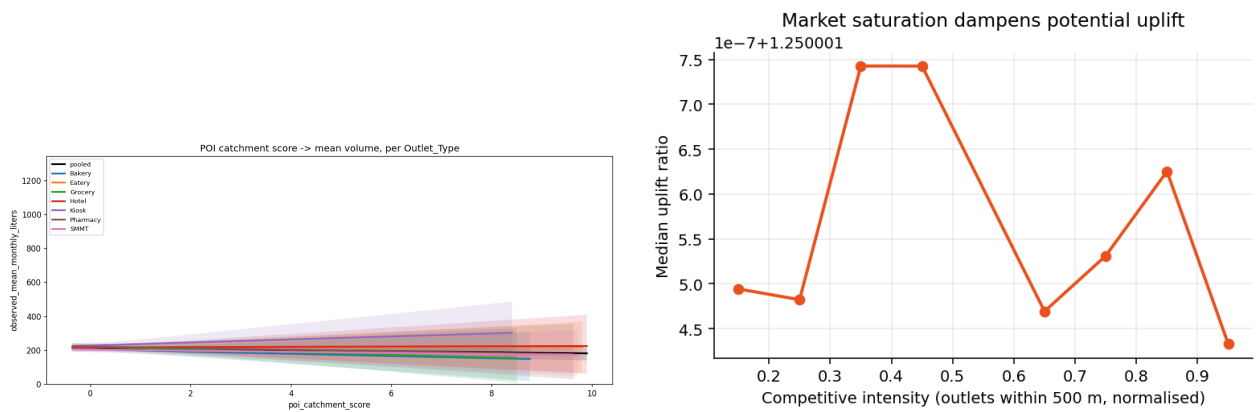
Spatial distance-decay (§2.1). Each category contributes a Gaussian-decay catchment score so a bus stop 20 m away dominates one 400 m away:

$$\text{score}_{i,c} = \sum_{p \in c} \exp\left(-\frac{d(i,p)^2}{2\sigma_c^2}\right),$$

with d the haversine distance and σ_c a category bandwidth; the nine scores are z-averaged into `poi_catchment_score`.

Competitive catchment density (§2.2). We count outlets within 500 m (`competitor_count_500m`) and 1/2/5 km, plus `same_distributor_outlet_count_5km`, `nearest_outlet_distance_km`, and `cannibalisation_count_200m` normalised into `competitive_intensity` ∈ [0, 1] separating a crowded cluster from an untapped market.

Honest disclosure. Per-outlet POI counts correlate only weakly with volume ($|r| \leq 0.026$ across `Outlet_Type` groups), so POI enters *only* via the composite score at ~20% weight in the constraint score – never as a standalone predictor. We flag the OSM-LK mapping bias: small kades and roadside stalls (our targets) are under-mapped.



(a) POI catchment is a *location* signal, not a direct predictor. (b) Denser competition $\Rightarrow$ lower median uplift (saturation).

Figure 2: External-data signals: POI catchment (left) and competitive saturation (right).

2 Data Cleaning

2.1 Initial quality assessment – six reusable checks

Function (checks.py)	What it catches
<code>duplicate_check(df, key_columns)</code>	duplicate composite keys
<code>null_check(df, columns)</code>	NaN / empty mandatory fields
<code>range_check(df, col, min, max)</code>	numeric out-of-range
<code>domain_check(df, col, allowed)</code>	misspellings, unexpected enums
<code>referential_integrity_check(...)</code>	foreign keys not in master
<code>geospatial_bounds_check(...)</code>	coordinates outside Sri Lanka

Table 3: All six DQ functions are parameterised and reused across datasets.

2.2 Programmatic cleaning & artifact neutralisation

Artifact	Records	Action
Outlet_Type typos + Outlet_Size case/missing	1,581	normalised / Unknown
Coordinates outside Sri Lanka (5.5–10°N)	240	quarantined
Non-positive Volume_Liters / Bill_Value	9,606	quarantined
Holiday duplicates (date+name+type)	93	deduped to rejected

Table 4: Every quarantined row carries a documented `failure_reason` (`src/cleaning/silver.py`).

2.3 Capacity from a measured saturation knee

Volume *stops growing* after 3 coolers (notebook 23): median monthly volume at 3, 4 and 5+ coolers is statistically indistinguishable. We use `Cooler_Count` as the deterministic sign anchor for the PCA capacity component – a physical representation of the cooler-replenishment ceiling.

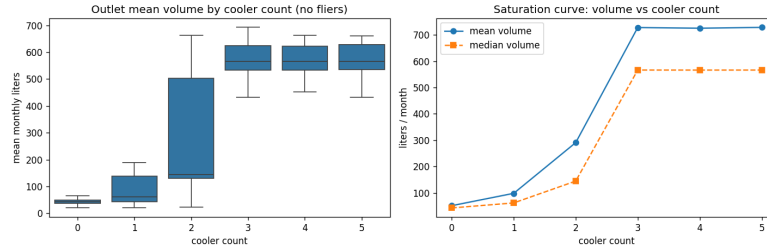
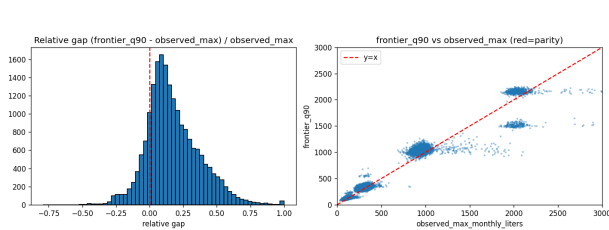


Figure 3: Saturation knee at 3 coolers anchors the capacity feature.

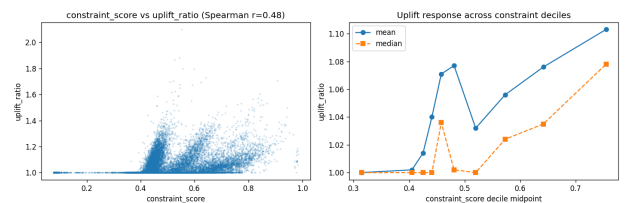
3 The Mathematical Framework

3.1 Identification: right-censored, not point-identified

The estimand $\mathbb{E}[\text{true_demand}_i \mid X_i]$ is **not point-identified** from observational data (no exogenous variation in constraints). $y_i = \min(\text{true_demand}_i, \text{constraint}_i)$ is right-censored – for any constrained outlet we see only a lower bound. We report a *band* alongside the point. For 15.62% of outlets the q90 frontier lies *below* observed historical max; without a $\max(\cdot, \text{obs_max})$ floor, validation V3b fails for those rows.



(a) Negative frontier gap motivates the `obs_max` floor.



(b) Constraint score drives per-outlet uplift, capped by evidence.

Figure 4: Censoring evidence (left) and the constraint-to-uplift mechanism (right).

3.2 Method stack

Robust lower bound + frontier ensemble. $\text{lower}_i = \max(\text{3rd-highest month}_i, \text{median}_i)$. The frontier F_i is a 60/40 blend of (a) **XGBoost 2.0** with `reg:quantileerror`, `quantile_alpha=[0.5,0.75,0.9,0.95]`, monotone constraints + isotonic post-sort, and (b) **SFA**: $y = X\beta + v - u$ with truncated-normal u , direct MLE; per-outlet $\text{TE} = \exp(-\mathbb{E}[u | \varepsilon])$ ($\sigma_v \approx 0.41$, $\sigma_u \approx 0.78$, $\lambda \approx 1.92$; median $\text{TE} \approx 0.84$). Leaky `observed_*` aggregates are dropped from the SFA design matrix.

Censoring correction + constraint score. Chernozhukov–Hong (2002) 3-step: a propensity $P(\text{censored} | X)$ on plateau proxies, then q90 refit on $P < 0.10$ (only 1.16 % of outlets show a true plateau – a small-population correction). The constraint score $s_i \in [0, 1]$ blends three *orthogonal* signals: peer-q90 frontier residual z-score, plateau gate, PCA-decorrelated capacity.

Final formula + calibrated interval.

$\text{potential}_i = \max(\text{obs_max}_i, \text{lower}_i + s_i(F_i - \text{lower}_i), \mathbf{1}_{[s_i \geq 0.4]} \cdot 1.25 \cdot \text{obs_max}_i)$ capped at $B_i \cdot \text{obs_max}_i$,

where B_i is the empirical 95th-pct uplift inside outlet i 's (`Outlet_Type`×`Outlet_Size`) bucket. We additionally report a calibrated $[q_{05}, q_{95}]$ via **Conformalised QR** on a 20 % outlet-level holdout, plus **Manski** worst-case bounds.

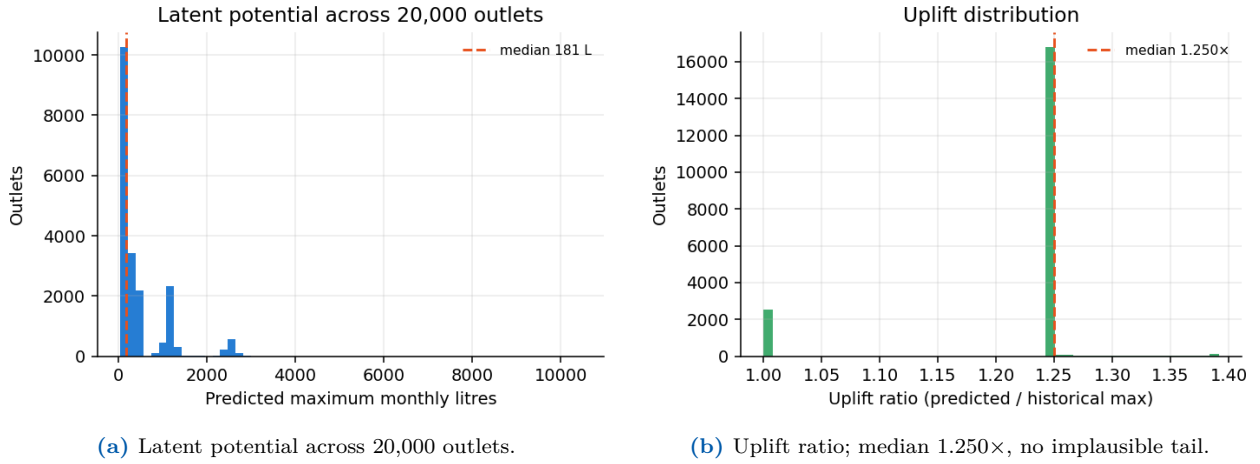


Figure 5: Prediction outputs: potential (left) and uplift distribution (right).

4 Spend Optimization Logic

Problem. Distribute a fixed **LKR 5,000,000** Western budget across the 9,000 Western outlets (`DIST_W_01/02/03`) to *maximise additional sales volume* over the normal baseline, with total spend $\leq$ budget. Western holds the largest untapped headroom of the four provinces.

Response model (concave, saturating). Incremental litres from spend s : $g_i(s) = H_i(1 - e^{-s/k_i})$, where H_i is latent headroom and the responsiveness scale grows with revenue value and competition: $k_i = \alpha(H_i \cdot \text{bill_per_litre}_i)(1 + \beta \text{comp_intensity}_i)$. A revenue-anchored cap $\text{cap_frac} \cdot H_i \cdot \text{bill}_i$ prevents a few large outlets from absorbing the budget ($\alpha=0.15, \beta=0.50, \text{cap_frac}=0.35$).

Exact solution – Lagrangian water-filling (KKT). Since $\sum_i g_i(s_i)$ is separable-concave, the optimum is exact. Stationarity gives $g'_i(s_i^*) = \lambda$, and with $g'_i(s) = \frac{H_i}{k_i} e^{-s/k_i}$ each spend is closed-form in the shadow price: $s_i^*(\lambda) = [k_i \ln(\frac{H_i}{k_i \lambda})]_0^{\text{cap}_i}$. We **bisect on** λ until $\sum_i s_i^* = B$ (budget-feasible, optimal) – not a heuristic ranking.

Metric (Western, LKR 5M, Jan 2026)	Value
Budget utilisation	98.0 % (LKR 4,901,675)
Outlets funded	3,729 / 9,000
Expected incremental volume	+109,021 L
Expected incremental revenue	LKR 27.08 M
Litres per LKR 1,000 (blended)	21.8 L
Revenue-to-spend ROI	5.53×

Table 5: Results/budget_allocation_summary.json. Submission: `smile_labs_budget_allocations.csv`.

The allocation concentrates spend on **high-headroom, high-margin, moderately-competitive** outlets – most untapped potential per rupee – rather than rewarding saturated top sellers or starving isolated shops.

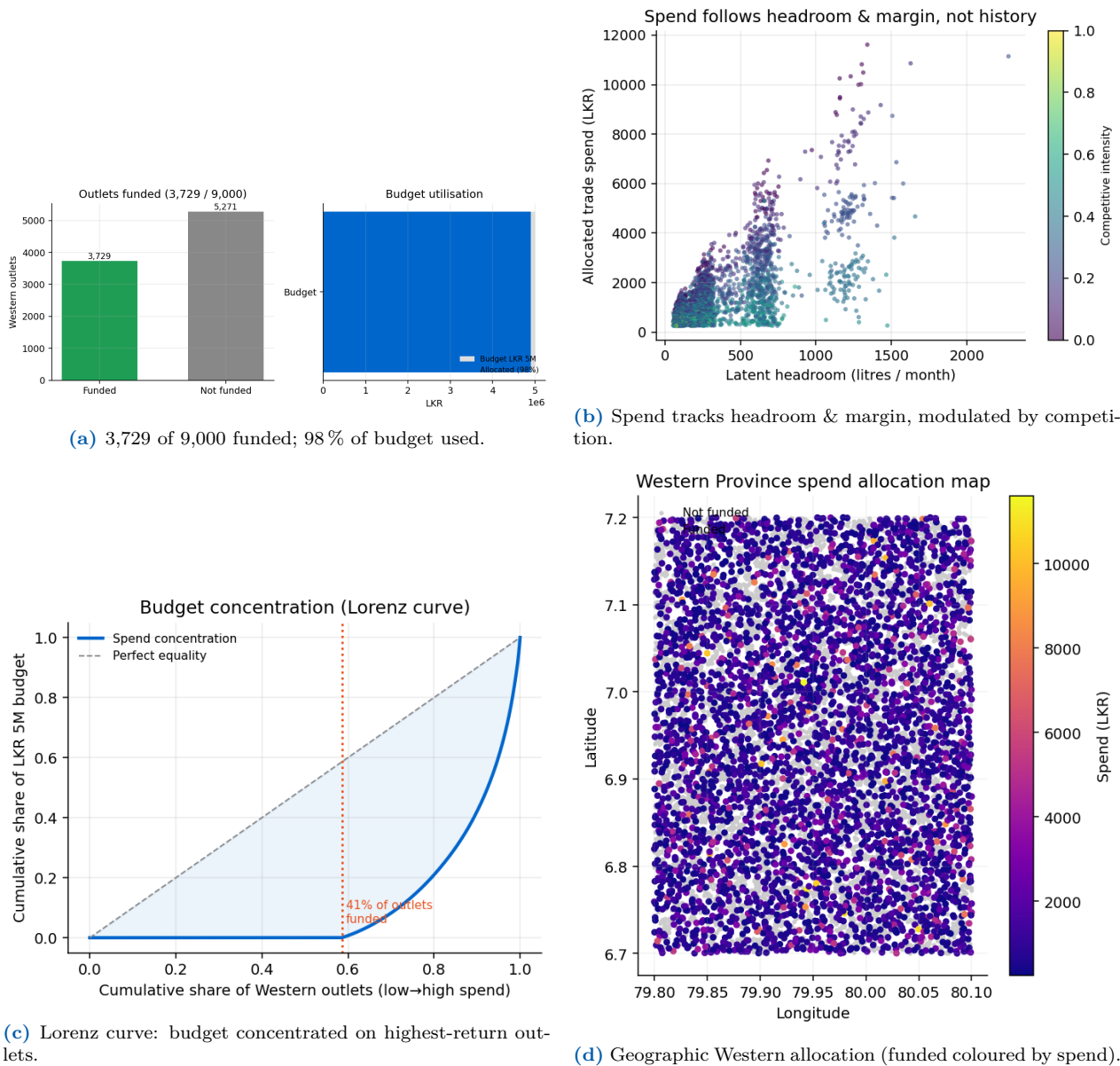


Figure 6: Spend optimization outputs: utilisation, mechanism, concentration, and geography.

5 Functional Explainable AI (XAI)

The XAI module (`src/xai/`) is a **user-facing layer** embedded in the web app, in two stages mapped to the four signal groups the brief requires.

1. **drivers.py – signed driver payload.** Converts a prediction into a ranked, signed set of drivers: *model drivers* (latent headroom, best-month anchor, history depth), *local signals* (competitive density, POI intensity), and *operational constraints* (cooler capacity vs the size norm). Each carries a direction, a relative weight, and a one-line technical detail.
2. **narrative.py – LLM narrative.** An LLM translates the payload into plain business language (why this score, which factors moved it, how local conditions shaped it). Uses the **Anthropic API** (`claude-opus-4-8`) when `ANTHROPIC_API_KEY` is set, with a deterministic **offline fallback**. Both paths consume the same structured payload, so narratives are validated against the drivers, never free-floating (samples in `Results/xai_samples.json`).

6 Validation, Outputs & Web App

6.1 Validation without ground truth (6-item checklist)

#	Check	Detail (final run)	Status
V1	schema + row count	[Outlet_ID, Max_Monthly_Liters], 20,000	OK
V2	no NaN/neg, unique IDs	NaN=0, neg=0, unique=True	OK
V3a	every ID in outlet_master	missing = 0	OK
V3b	pred $\geq$ hist. max ($\geq 99\%$)	0.00 % below	OK
V4	median uplift $\in [1.25, 2.2]$	median = 1.250	OK
V5	cap-binding < 25%	0.00 % at cap	OK

Table 6: src/reporting/validation.py. Sensitivity sweep across quantile $\times$ scheme $\times$ cap keeps median uplift at $1.250\times$.

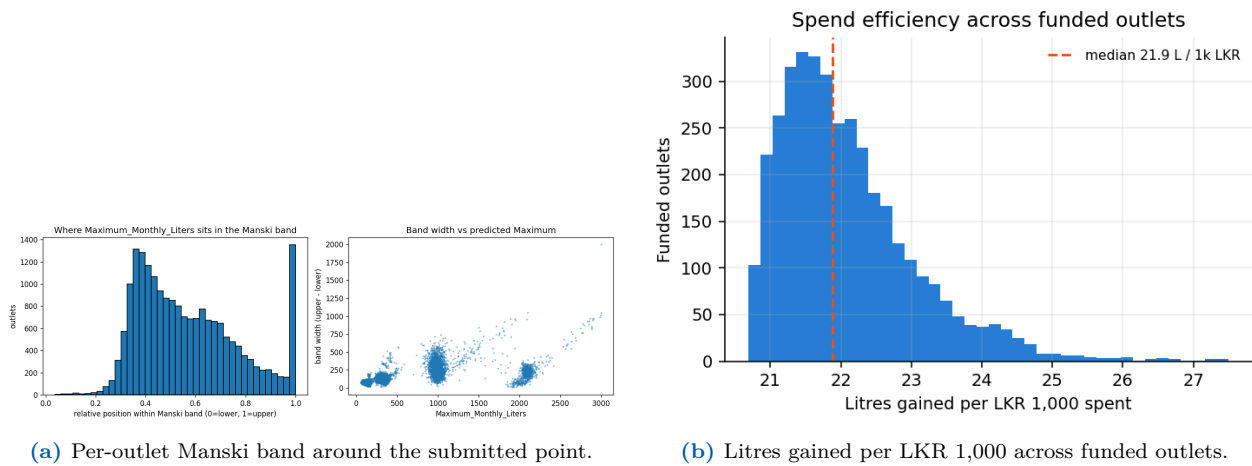


Figure 7: Uncertainty disclosure (left) and spend efficiency (right).

6.2 Outlet Intelligence web app

A local **Streamlit** app (app/streamlit_app.py, no external services) with three tabs: **Browse** (filter by province/distributor/type, search, map, CSV export), **Drill-down** (per-outlet metrics, signed drivers, on-demand XAI narrative with live/offline toggle, raw payload), and **Western spend plan** (the LKR 5M allocation with utilisation, funded count, expected uplift, and submission download).

7 GenAI Transparency Log

Phase	AI usage	Human validation
Research swarm	10-channel parallel research swarm $\rightarrow$ research brief	each claim cross-checked vs cited URLs
Methodology audit	7 rounds of 4–5 parallel critics (Statistician, Skeptic, Architect, DE, Gap, Judge, Risk)	every finding cites file:line; verified / rebutted
Code + design	JLMS closed form; OSM tag survey; Geofabrik vs Overpass; KKT derivation	syntax / import checks; SFA vs Greene; optimizer proven concave
XAI integration	narrative prompt engineering; driver schema design	narratives validated vs structured drivers; offline tested
Report drafting	LaTeX assembly + content port from markdown	every claim traced to Results/ or data/gold/

Table 7: Per-phase GenAI transparency. Full log: Docs/ai_transparency_log.md (R1–R7 audit trail).

Deliverables

Artifact	Path
Platform predictions CSV	Results/smile_labs_predictions.csv (20,000 rows)
Budget allocations CSV	Results/smile_labs_budget_allocations.csv
Manski + conformal bands	Results/manski_bands_v2.csv, conformal_intervals_v2.csv
Validation	Results/validation_report.md (6/6 PASS)
Outlet Intelligence app	app/streamlit_app.py
Reproducible codebase	run_pipeline.py + run_final_round.py; src/

References

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